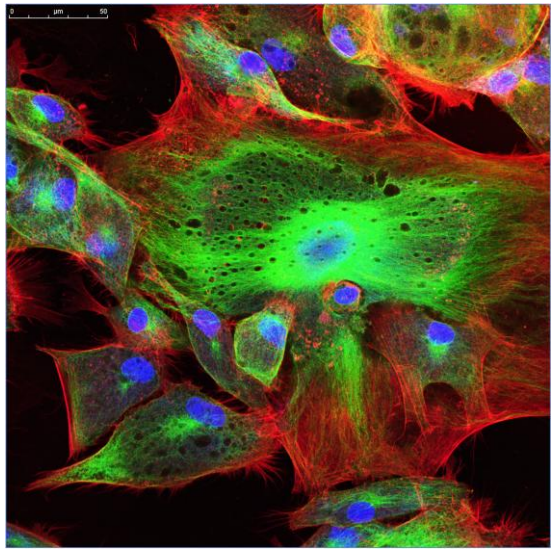
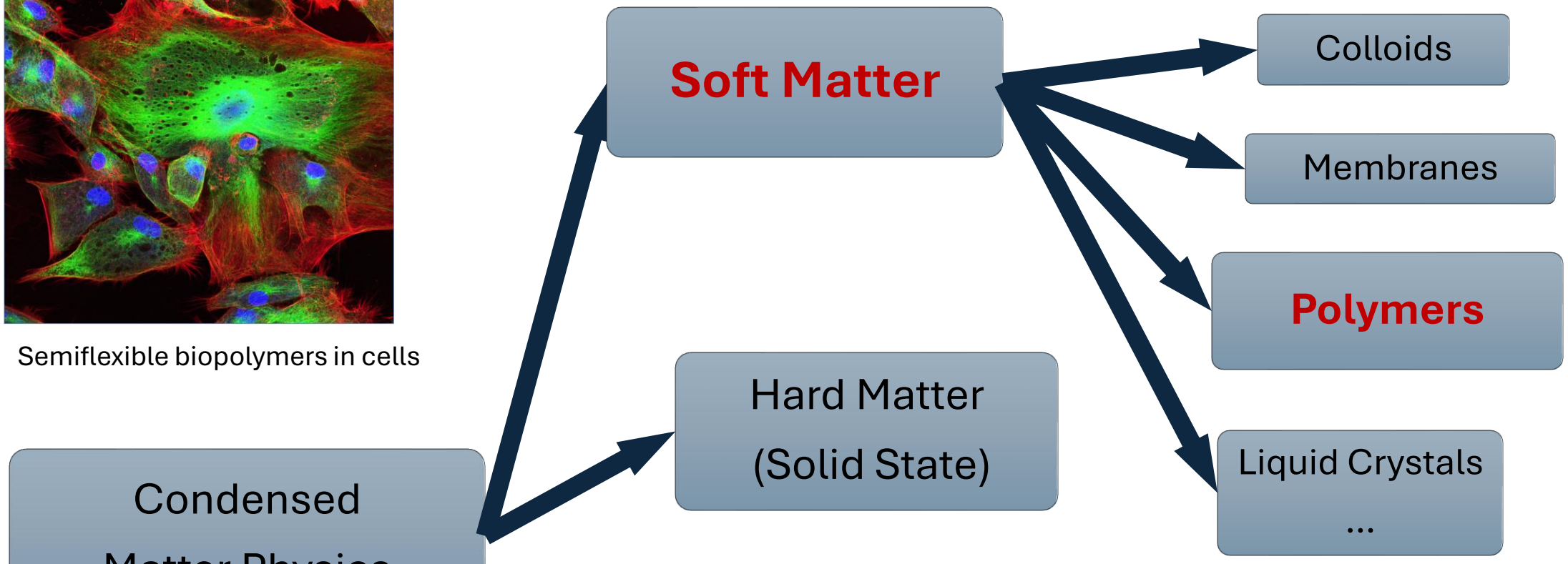
Soft Matter Theory Lab



Semiflexible biopolymers in cells



What we are doing

- Motivated by specific systems in **soft/biological matter**, we use the analytical tools of **Statistical Mechanics** to gain understanding of **fundamental** aspects of the behavior of those systems. Typical questions that we address:

- How does the topology of polymers (e.g., loops vs strings) affect their elasticity?
- Can we get order out of disorder (e.g., introduce permanent random cross-links in a polymer brush and get polymer bundles)?
- What are the microscopic limits of thermodynamics at the molecular level?
- How is transport of molecular motors on biopolymers affected by the polymer fluctuations?

Members:

- Prof. Panayotis Benetatos
- Geunho Noh (PhD student)
- Minsu Yi (Master's student)

Recent publications:

- M. Yi and P. Benetatos, Elasticity of a Freely Jointed Chain with Quenched Disorder (under review) arXiv preprint arXiv:2503.14817
- M. Yi, D. Lee and P. Benetatos, Bending Elasticity of the Reversible Freely Jointed Chain, *J. Chem. Phys.* **161**, 234908 (2024)
- G. Noh and P. Benetatos, Stretching Multistate Flexible Chains and Loops, *Phys. Rev. E* **110**, 014501 (2024)